

# Harry Nguyen

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## EDUCATION

### Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

## EXPERIENCE

### TiMoto AI | Software Engineer

Sep 2025 – Present

- Built and operated **distributed production systems** end-to-end as primary engineer on 3-person team – backend, cloud infra, and ML serving; participated in **on-call rotation** and maintained runbooks from every incident
- Diagnosed production deadlock under concurrent **gRPC** calls – traced circular resource acquisition order across shared handler state and redesigned call sequencing; restored **100% evaluation success rate** at **sub-50ms p99**
- Architected **vLLM** inference engine with **PagedAttention** – eliminated KV cache memory fragmentation under concurrent load via continuous batching; **zero OOM failures** at production traffic
- Designed multi-AZ ECS Fargate topology with **Terraform IaC**, CloudWatch observability, and circuit breaker pattern – **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure
- Maintained **geographically-dispersed** service layer across gRPC, PostgreSQL, and ML serving – triaged incidents via root-cause analysis and applied post-mortem learnings to harden call-path correctness

### Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Eliminated **mutex contention** via **lock-free concurrent trie** search on Chrome settings navigation – **96% latency reduction** (1,200ms to ~50ms p99), zero production regressions; verified linearizability before shipping
- Designed **C++ IPC transport layer** with **Protocol Buffers** – selected Protobuf over custom binary format for schema evolution and cross-language compatibility; shipped to Chrome stable channel serving **3B+ users** at **sub-50ms p99**, **10K+ req/sec**
- Architected event-driven TypeScript/React system with observer pattern – accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**
- Delivered design documents and code reviews reviewed by senior Chrome infra engineers; changes adopted into production branch

### Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Diagnosed **N+1 query bottleneck** degrading response to 3s+ on large datasets; redesigned with **PostgreSQL** composite indexing achieving **sub-100ms** for 10,000+ records
- Designed stateless BaaS on **AWS** – JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via **CI/CD** (GitHub Actions)

## PROJECTS

### Pulumi – Contributor | Go, TypeScript, IaC

[github.com/pulumi/pulumi](https://github.com/pulumi/pulumi)

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

## SKILLS

**Distributed Systems:** gRPC, Protocol Buffers, Semaphores/Mutexes, Lock-free structures, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

**Systems & OS:** C++, IPC, Lock-free concurrency, Filesystems, OS internals, Compiler toolchain, Network protocols, Raft/Paxos

**Languages:** C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

**Cloud & Infrastructure:** AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP

**ML & AI Infrastructure:** vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation

**Frameworks & Databases:** Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

**AI Dev Tools:** Claude Code, GitHub Copilot, Codex, Cursor